FlatSplit

Evidence-Backed Group Rental Search

CS5224 Cloud Computing - Group 12

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Contents

Abstract

1. Motivation and Business Case

2. Architecture and Implementation

3. Evaluation

4. Conclusions and Future Work

5. Code Explanation and Reproduction

6. Team Contributions

7. AI-Assistance Declaration

References

> Review status: integrated evidence draft; latency/capacity, participant outcomes and final qualification remain explicitly pending.

Abstract

FlatSplit is a serverless web service for groups of two to five people choosing an HDB rental town in Singapore. Existing rental portals optimise for an individual listing search; they do not combine every housemate's budget, destination and maximum commute into one explainable group decision. FlatSplit uses official HDB rental data and OneMap public-transport routes to filter infeasible towns and rank the remainder by rent per person, average commute and commute inequality. The implementation uses CloudFront, S3, API Gateway, Lambda, DynamoDB, SQS, EventBridge, Secrets Manager and CloudWatch. Its central cloud design choice is to keep interactive requests short while moving provider-bound routing and monthly data publication into asynchronous, bounded workflows.

The evidence supports the core data and reliability claims. A rental audit matched 65 of 72 numeric HDB quarterly medians exactly; median absolute error was S$0 and maximum error was S$100 (2.70%). A reviewed route audit covered 810 town-destination pairs: all 780 routable pairs succeeded and 30 TENGAH pairs were explicitly unavailable rather than represented as zero minutes. A deployed drill showed three bounded deliveries followed by dead-lettering and a CloudWatch alarm. The cost model estimates typical serverless cost at US$0.96/month under assumed allowances, against US$5.71 at list price, US$243.50 for self-managed on-premise hosting, and US$85.94 for a university-hosted VM. However, the cost figures are modelled rather than billed, station snapping systematically understates door-to-door commutes, deployed transient recovery was not induced, and the latency, capacity and participant study results remain pending at this draft cut-off.

1. Motivation and Business Case

Singapore students and young professionals often share an HDB flat to reduce housing cost. The difficult decision is not merely which listing is cheapest, but which town is acceptable to the whole group. Housemates may work or study at different destinations, have different budgets, and tolerate different commute lengths. An apparently good average can conceal one member's unacceptable journey.

FlatSplit addresses that coordination problem at town level. Each member owns their destination, monthly budget and maximum commute input. Once every member is ready, the service returns feasible towns, per-person rental benchmarks, each member's estimated commute, the longest journey and the gap between the shortest and longest journeys. Excluded towns remain visible with specific reasons such as over budget, over commute, low-confidence rent or unavailable route. The prototype does not provide individual listings, room allocation, eligibility advice, payments, bookings or a guarantee that a flat is available at the benchmark price.

1.1 Target users and value proposition

- **Primary users:** groups of two to five students or young working adults renting together in Singapore.
- **Problem solved:** combine multiple personal constraints into one reproducible, explainable comparison instead of coordinating through screenshots and repeated messages.
- **Differentiation:** existing portals search listings; FlatSplit compares group-level town trade-offs and exposes the most burdened member rather than hiding them inside an average.
- **Potential customers:** university communities, relocation services and property professionals supporting shared households.

1.2 Business model

The prototype is a free decision-support service. A future freemium product could charge for saved searches, rental trend alerts, scenario history or professional comparison reports. Distribution could use university housing communities, referrals and relocation partnerships. These revenue features are not implemented and are not needed to validate the technical proposition.

The workload is seasonal and bursty: searches cluster around semester and relocation periods, while the service may be almost idle between them. This favours request-priced serverless components over continuously running capacity. The corrected cost model compares four like-for-like options:

Hosting option	Quiet US$/month	Typical US$/month	Bursty US$/month	Important boundary
AWS serverless, modelled payable	0.82	**0.96**	**2.09**	Assumed allowances; not an observed bill
AWS serverless, list price	5.11	**5.71**	**10.45**	Free-tier credits removed
AWS always-on VM	38.32	**38.44**	**56.92**	Same application on fixed capacity
Self-managed on-premise	243.50	**243.50**	**274.87**	Likely effort; range US$188.81-415.38
University-hosted VM	85.94	**85.94**	**85.94**	Infrastructure absorbed; range US$31.25-257.81

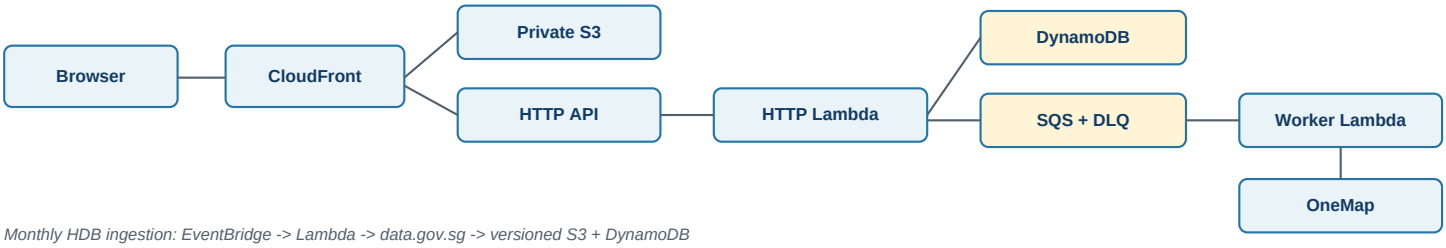
At typical demand, even serverless list price is about 43 times lower than the self-managed on-premise estimate. The strongest result is the idle shape: US$0.82 for serverless versus US$243.50 for self-managed hosting. At this scale, fixed operational commitments dominate. Two Secrets Manager secrets dominate the modelled payable cloud total; people and connectivity dominate the on-premise total. The conclusion remains conditional because the AWS account's cost-allocation tag was not available to the linked account, applicable credits are unconfirmed, and `sustained_rps_per_node` remains assumed until #30 supplies a capacity measurement [1].

2. Architecture and Implementation

2.1 Implemented user journey

1. An organiser creates a two-to-five-person group and receives an invitation URL. The capability is carried in the URL fragment so it is not sent in HTTP requests or server logs.
2. Each member joins in an independent browser context, explicitly selects a OneMap destination, and saves a budget and maximum commute.
3. The API persists each member's input and advances the group's revision. A search is enabled only when all declared members are ready.
4. Starting a search pins the group revision, rental snapshot and routing profile. Cached commutes are reused; missing station-town pairs are claimed and queued.
5. The frontend polls real progress. When all pairs are resolved, it displays ranked and excluded towns, individual outcomes, side-by-side comparisons and an accessible reference-point map.
6. A later preference change increments the revision and withholds stale results until the group searches again.

2.2 Cloud architecture



Monthly HDB ingestion: EventBridge -> Lambda -> data.gov.sg -> versioned S3 + DynamoDB

CloudFront provides one browser-facing origin: static requests go to a private S3 bucket through origin access control, while `/api/*` goes to API Gateway with API caching disabled. Group, place and search routes share one Python 3.13 arm64 HTTP Lambda rather than the three logical API Lambdas proposed in the preliminary design. This lowers deployment overhead and keeps the prototype simple while preserving module boundaries in code.

The HTTP path stores state in one on-demand DynamoDB table. SQS isolates slow, quota-bound route work from the interactive request. Each message represents one canonical station-town pair, so aliases and repeated searches share cached results. The worker processes one message per invocation with event-source maximum concurrency three. A fixed-window DynamoDB counter reserves up to 200 routing calls/minute and leaves 50 of the shared 250-call budget for interactive destination lookup. This guard is implemented and locally tested; the deployed warm-up reached 125 calls/minute sustained without triggering it, so deployed enforcement at the ceiling remains unmeasured.

2.3 Security and state consistency

The implementation intentionally has no login wall. It uses separate random invite and member capabilities; only SHA-256 hashes are stored. Invite capabilities can join but cannot read the group. Member capabilities are scoped to their group and expiry. Browser-held capabilities are excluded from publishable logs and evidence.

IAM roles are separated by workload. The HTTP Lambda can access only required table prefixes, its queue and token secret. The commute worker can update only `COMMUTE#*` items, consume its queue and read the OneMap token. The ingestion Lambda writes versioned rental evidence and rent partitions. S3 buckets are private and encrypted; SQS denies non-TLS access. DynamoDB point-in-time recovery is enabled, although the E6 restore drill has not yet been executed.

Conditional writes protect correctness under retries. Group revisions prevent old searches from surviving changed preferences. Commute completion requires a matching `jobId` and `pending` status, so duplicate or delayed messages cannot overwrite a newer claim. Rental summaries are written under immutable snapshot IDs before the `RENT/CURRENT` pointer advances. Search records pin their rental snapshot and routing profile, so a completed result cannot silently change while a user is reading it.

2.4 Rental and commute data

FlatSplit ingests HDB's cumulative **Renting Out of Flats from Jan 2021** export from data.gov.sg [2]. The profiled September 2026 export contains 212,597 rows, 69 continuous source months, 27 towns and six flat types. Its six source fields are approval month, town, block, street, flat type and monthly rent. FlatSplit interprets monthly rent as SGD per whole flat per month; the source metadata does not state the unit, so this remains an explicit project assumption.

The monthly ingestion validates the exact schema, town and flat-type inventory, positive integral rents, complete trailing-12-month coverage and consistency with source metadata. Accepted files are archived in versioned S3 paths with a digest and manifest; rejected evidence never advances the current pointer. For every non-empty town-flat-type group, the publisher stores R-7 P25, median and P75 plus a deterministic 10,000-replicate block-cluster-bootstrap 95% interval. A group is usable only when it contains at least two independent blocks and the median interval width is at most 10% of the median. Weak evidence remains visible but cannot affect affordability or ranking.

Commutes use OneMap public-transport routing [3]. Each destination is snapped to a reviewed, pinned MRT station within 1,000 m, with a warning above 800 m. Each town is represented by a principal station. Routes use weekday 08:30 departure, request three itineraries and retain the shortest. These are station-to-station estimates, not door-to-door journeys. TENGAH is retained as an explicit non-routable town rather than removed from the dataset.

2.5 Feasibility and fairness ranking

Group size maps to an indicative whole-flat type: two members to 3-room, three to 4-room, and four or five to 5-room. Rent per person is the town median divided equally by group size; utilities and room-specific allocation are excluded. This mapping is a prototype policy, not HDB occupancy or eligibility advice.

A town is feasible only when its rent summary is usable and every member meets both constraints:

```
rentPerPerson <= member budget
commuteMinutes <= member maximum commute
```

Feasible towns receive a transparent demonstration score:

```
cost = wRent * rentPerPerson/2000
     + wAverage * averageCommute/90
     + wGap * (longestCommute - shortestCommute)/90
score = max(0, round(100 * (1 - cost)))
```

The presets are balanced `(0.35, 0.35, 0.30)`, fairness `(0.20, 0.25, 0.55)` and budget `(0.70, 0.20, 0.10)`. Changing the preset re-ranks the same pinned evidence without starting a new search. The score is an inspectable index, not a satisfaction probability. Unknown or unavailable data never becomes zero; excluded towns carry member-specific reasons.

3. Evaluation

Evidence is versioned with scripts, raw output, dates and limitations under `docs/evaluation/`. The table separates deployed measurements from local or modelled evidence.

Dimension	Evidence at 8 Nov	Result	Boundary
Rental quality	Source audit and deployed summaries	65/72 exact numeric medians; median absolute error S$0; 75/81 recommendation groups usable	Both datasets are HDB sources, not independent market truth

Dimension	Evidence at 8 Nov	Result	Boundary
Route coverage	Live OneMap audit + deployed reconciliation	780/780 routable pairs; 30/30 TENGAH pairs unavailable; 810/810 status agreement	Station-to-station, one weekday departure
Snapping accuracy	90 address-vs-station comparisons	Median absolute error 4.1 min; maximum 16.8; 82/90 understated	Three alphabetically selected probe towns; not random
Failure containment	Deployed `eval` drill	Three deliveries, DLQ, alarm, no loop; permanent rejection delivered once	Transient provider recovery not induced
Operational clarity	Independent #33 walkthrough	Correct cache/log/DLQ inspection and no-redrive decision	One teammate and one permanent-unavailable case
AWS usage	124-hour shared-dev capture	763 HTTP requests, zero 5xx/Lambda errors; US$0.03377 variable list-cost estimate	Not a controlled representative run or bill
Hosting cost	Reproducible scenario model	US$0.96 typical serverless payable vs US$243.50 on-premise	Demand, account allowances and node capacity partly assumed
Latency/capacity	**Pending #30**	No accepted W1/W2 result at this cut-off	500 ms p95 remains a target, not a result
User outcomes	**Pending #31**	Protocol and fixtures exist; no participant result integrated	Local QA is not a usability study

3.1 Rental-number audit

The Q2 2026 audit compared medians calculated from detailed HDB rental records with HDB's separately published quarterly town/flat-type medians [4]. Of 81 possible 3/4/5-room groups, 72 had numeric published medians. Sixty-five matched exactly; the remaining seven differed by at most S$100 or 2.70%. Median absolute difference was S$0 and mean absolute difference S$3.40. Six published values were `-` and each corresponding detailed group contained fewer than 20 records; the marker was preserved as missing evidence rather than read as zero.

For the production trailing-12-month window, 75 of 81 scoped groups were usable, five were low-confidence and `TENGAH x 3-ROOM` was missing. The result justifies town-level benchmarks while demonstrating why a quality flag is necessary: `CENTRAL x 4-ROOM` had 204 rows but still failed the block-cluster stability rule.

3.2 Route coverage and approximation error

The reviewed catalogue contained 30 destinations and 27 HDB towns, producing 810 route pairs. Every pair for the 26 routable towns succeeded; all 30 TENGAH pairs were explicitly unavailable, with zero errors. Across 780 routed pairs, median commute was 33.0 minutes, p95 62.3 and maximum 82.4.

The deployed table contained 729 canonical station-town records because the 30 destinations mapped to 27 distinct stations. A strongly consistent reconciliation found a record for all 810 requested pairs and 810/810 status agreement. For the 780 successful routes, deployed-minus-audit travel time had median 0.00 minutes and p95 absolute difference 0.05 minutes. Twenty OUTRAM_PARK pairs differed by about +/-5.7 minutes, consistent with but not proving different points inside a multi-line interchange.

The approximation is materially biased. Across 90 exact-address-versus-snapped comparisons, median absolute error was 4.1 minutes, maximum 16.8 and 82 were negative. The station estimate systematically omitted the final walk. Median signed error grew from -2.4 minutes below 200 m to -11.9 minutes at 500 m or more. FlatSplit therefore labels results as station-to-station and exposes snap warnings rather than claiming door-to-door accuracy.

3.3 Failure recovery and operations

On 7 November, an isolated `eval` drill injected an unkeyable commute message. The worker received it exactly three times at the configured 180-second visibility interval, after which SQS moved it to the DLQ. The message was dead-lettered 9 minutes 37 seconds after injection and the `CommuteDlqNotEmpty` alarm entered `ALARM`. A separate keyable message with invalid coordinates was consumed once, logged as terminal and not redelivered. Cleanup deleted only drill messages by receipt handle; both queues returned to zero and all alarms returned to `OK`. The live event-source mapping was later read back as enabled with `MaximumConcurrency: 3`.

This establishes bounded failure and visibility, not recovery from a transient OneMap outage. W4 required controlled upstream fault injection, but the deployed stack has no fault switch and the team chose not to add failure-only code to the shared evaluation stage during the release week. Redrive was demonstrated only as a mechanism: replaying a permanently invalid message failed again. The token-warning alarm was not deliberately triggered because `eval` and `dev` share the provider account. The report therefore makes the narrower claim that the system does not loop and does not invent commute times; provider recovery is supported by code and local tests but unmeasured on AWS.

An independent teammate followed the operations runbook against `FlatSplit-eval`. The walkthrough found 26 successful town routes and TENGAH explicitly unavailable, confirmed null rather than zero commute values in a complete three-member search, located a successful worker event, found an empty DLQ and correctly decided not to redrive permanently non-routable TENGAH [5].

3.4 Performance and scalability

The architecture bounds routing concurrency and caches canonical pairs, but an admission limit is not a capacity result. The prepared W1 workload offers two complete-result reads/s for 10 minutes; W2 ramps from one to 50 requests/s and holds 50 requests/s for five minutes. As of this draft, two attempted runs failed on the load client before producing an accepted backend measurement. Therefore the 500 ms warm-result p95 objective, sustainable HTTP throughput and VM node sizing remain unverified. #30 is scheduled to supply either a valid result or an explicit final limitation.

The deployed route-catalogue warm-up provides a narrower observation: 702 real routing calls completed over approximately five and a half minutes, about 125 calls/minute sustained with a rolling peak of 136/minute, all on attempt one. No rate-budget refusal occurred. This shows headroom at the measured load, not enforcement once the worker's 200-call window is full.

3.5 User experience and accessibility

Local browser acceptance exercises two- and five-member groups in independent Chromium contexts at desktop and 390 x 844 mobile viewports. It covers group creation/join, explicit destination selection, validation, cross-member reload, asynchronous results, three-town comparison, keyboard map interaction, stale result removal and automated WCAG A/AA checks. The test uses the real local HTTP resolver and persistence, but synthetic rental/place records and injected commute completions. Automated accessibility checks are not a screen-reader or physical-device certification, and no participant outcome is claimed. The #31 study must add task completion, interpretation and qualitative findings or be reported as not run.

3.6 Cost evaluation

The latest cost model combines measured request shapes, deployed memory and duration observations, item sizes, log volume and scheduled-function work with explicit demand and hosting assumptions [1]. A 124-hour shared-dev observation recorded 763 API requests, 130 worker invocations, four rent-ingestion runs and 80 token-refresh runs. It observed zero API 5xx, zero HTTP/worker/ingestion Lambda errors and no throttles; six historical token-refresh errors predated the latest stack update. The measured variable units corresponded to approximately US$0.03377 at gross list price, excluding fixed and unmeasured lines.

The monthly comparison is a model, not an invoice. Cost Explorer could not filter `project=flatsplit` from the linked account, applicable credits are unconfirmed, and the on-premise model assumes hardware prices, electricity, exchange rate and operational effort. Capacity-sensitive self-hosted node counts still use an assumed 25 requests/s/node. The strength of the result is its sensitivity: even removing free-tier allowances, typical serverless list price remains far below both self-hosted alternatives.

4. Conclusions and Future Work

FlatSplit demonstrates that official rental and transport data can support an explainable town-level group decision service. The system does not hide unavailable evidence, weak rental samples or one member's constraint inside a single average. Versioned snapshots, canonical route caching, revision fences and bounded asynchronous work make the design suitable for a low-average, bursty SaaS workload. The cost model supports serverless hosting for this usage shape even without relying on free-tier credits.

The evidence also narrows the product claim. Rent figures are owner-declared historical benchmarks, not listings. Commutes are station-to-station and systematically omit the final walk. TENGAH remains visible but unrankable for commute comparison. Recovery from a deployed transient provider outage, the rate guard at saturation, PITR restoration, accepted latency/capacity, and participant understanding remain unmeasured at this cut-off.

Priority future work is to:

1. add station-to-address walking time or route to exact destinations while preserving cache sharing;
2. complete W1/W2 and use measured throughput to update self-hosted sizing;
3. add a safe, isolated fault-injection path and run transient recovery and PITR drills;
4. run the prepared participant study and refine weights based on whether users understand fairness trade-offs;
5. obtain payer-account cost allocation and compare the model with an actual tagged bill; and
6. replace the prototype group-size flat-type mapping with an explicit housing policy and user choice.

5. Code Explanation and Reproduction

The public repository is <https://github.com/horiiiiii032929/CS5225-teams12-2026>. It is a pnpm/uv monorepo:

Path	Responsibility
`apps/web`	React, TypeScript, Vite, TanStack Router/Query user interface
`services/api`	Python Lambda handlers, storage, validation, ranking and workers
`packages/contracts`	Generated public TypeScript API schemas
`infra/aws`	AWS CDK stack for web, API, data, queues, schedules and alarms
`tools/load`	Privacy-safe W1/W2 workload generation and summarisation
`tools/evaluation`	Reproducible data, route, cost and acceptance checks
`docs/evaluation`	Dated procedures, raw evidence and interpretation

Core request flow is implemented in `services/api/src/flatsplit_api/handlers/http.py`. Group persistence and capability checks live in `groups.py`; search claiming/status in `searches.py` and `commute_jobs.py`; ranking in `ranking.py`; OneMap integration and station pinning in `onemap.py` and place storage; the asynchronous consumer in `handlers/commute_worker.py`; and rental validation/publication in the ingestion modules. React feature components consume generated contracts rather than reaching into backend source.

From a clean checkout, install Node 24, pnpm 11.0.9, Python 3.13 and uv, then:

```
pnpm install --frozen-lockfile
uv sync --locked --all-packages --group dev
pnpm check
pnpm test
pnpm build
pnpm synth
```

`pnpm dev` starts the local frontend and API. The repository includes explicit instructions for deployed evidence; local emulators and synthetic browser fixtures must not be relabelled as cloud acceptance.

6. Team Contributions

This draft attribution is based on issue ownership, merged pull requests and the contribution packets in #38. Every member must confirm or correct it during the 9-12 November factual review.

Member	Main contributions
Dong Langqing (Lucy)	Member-input and results UI, mobile/accessibility QA, reproducible demo preparation and presentation coordination
Horinouchi Hikaru	Group creation/share/reopen frontend integration, project coordination, cross-cutting reviews and release/evidence tracking
Hsieh Ming Cheng	Destination resolution and station pinning, commute worker/reliability evidence, route-coverage evaluation, runbook and hosting-cost model
Ku Wee Kiat	Capability-backed group persistence, transactions and expiry; revision-aware search producer/status flow; backend verification and load-test preparation

Member	Main contributions
Ramalingam Saravanamani	HDB rental profiling, ingestion and validated summaries; data-quality and deployed-cost evidence review; AWS deployment checks and final-report integration
Shyam Ganesh Jayagopi	Group HTTP APIs, filtering/ranking integration, AWS infrastructure/deployment coordination, deployed E2E/operator receipts and evaluation support

7. AI-Assistance Declaration

The team used AI coding assistants for implementation support, review, documentation and evidence reconciliation. Team members directed decisions, reviewed generated changes, ran the reported checks and retained responsibility for the submitted work. The repository's `docs/ai-usage.md` records the tools, scoped contributions and verification. This integrated draft was assembled with Codex assistance from team-authored design documents, merged code, issue decisions and dated evidence; it introduces no new experimental result.

References

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